

Representations of a category can be those of a groupoid

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The general problem

- ▶ Let R be a commutative ring. Let $\mathcal{V} = \text{Ab}^R$ be the monoidal category of R -modules. For any category $\mathcal{F}$, let's call the functor category $[\mathcal{F}, \mathcal{V}]$ the *category of R -linear representations of $\mathcal{F}$* .
- ▶ I am interested in categories $\mathcal{F}$ for which there is a groupoid $\mathcal{G}$ such that the categories $[\mathcal{F}, \mathcal{V}]$ and $[\mathcal{G}, \mathcal{V}]$ of representations are equivalent.
- ▶ In particular, $\mathcal{G}$ could be the core groupoid $\mathcal{F}_{\text{inv}}$ of $\mathcal{F}$; that is, the subcategory with the same objects and with only the invertible morphisms.
- ▶ We could freely split some idempotents in $\mathcal{F}$ without changing the representations. However, adding the new objects may change the core groupoid.

Linearizing

- ▶ Every category $\mathcal{F}$ gives rise to a $\mathcal{V}$ -category (that is, an R -linear category), denoted $R\mathcal{F}$, with the same objects and with hom R -module $R\mathcal{F}(A, B)$ free on the homset $\mathcal{F}(A, B)$.
- ▶ Indeed, $R\mathcal{F}$ is the free $\mathcal{V}$ -category on $\mathcal{F}$ so that the $\mathcal{V}$ -functor category $[R\mathcal{F}, \mathcal{V}]$ is isomorphic to the ordinary functor category $[\mathcal{F}, \mathcal{V}]$ with the pointwise R -linear structure.
- ▶ In these terms, we are interested in when $R\mathcal{F}$ and $R\mathcal{G}$ are Morita equivalent $\mathcal{V}$ -categories so that a groupoid suffices for all linear representations of $\mathcal{F}$.
- ▶ Lack-St (2015) showed that proving a Dold-Kan-type equivalence of categories reduced to solving a “core groupoid suffices” problem.

An example without details

- ▶ Indeed the original Dold-Kan equivalence between simplicial R -modules and chain complexes of R -modules reduces to the “core groupoid suffices” problem for $\mathcal{F} = \Delta_{\perp}$.
- ▶ Here $\Delta_{\perp}$ is the category of finite non-empty ordinals

$$\mathbf{a} = \{0, 1, \dots, a - 1\}, \quad a \in \mathbb{N},$$

and order-and-first-element preserving functions $\xi : \mathbf{a} \rightarrow \mathbf{b}$.

- ▶ As the only invertible morphisms in $\Delta_{\perp}$ are identities, the core groupoid in this case is discrete with countably many objects; so the category of representations is the product $\mathcal{V} \times \mathcal{V} \times \dots$ of countably many copies of $\mathcal{V}$, that is, the category of objects of $\mathcal{V}$ graded by the positive integers.

A baby example with details, page 1

- ▶ Let $C = \mathbf{3} = \{0, 1, 2\} \in \Delta_{\perp}$. The monoid $\Delta_{\perp}(C) = \Delta_{\perp}(\mathbf{3}, \mathbf{3})$ has six elements, namely, the functions

$$\iota = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \quad \sigma_0 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \quad \sigma_1 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\tau = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} \quad v = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \zeta = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- ▶ A presentation of this monoid is provided by the generators σ_0 , σ_1 and τ subject to the relations that the generators are idempotents and

$$\sigma_0 \sigma_1 \sigma_0 = \sigma_0 \sigma_1 = \sigma_1 \sigma_0 \sigma_1$$

$$\tau \sigma_0 = \sigma_0, \tau \sigma_1 = \tau = \sigma_0 \tau, \sigma_1 \tau = \sigma_1.$$

A baby example with details, page 2

- ▶ The only invertible element in $\Delta_{\perp}(C)$ is the identity ι .
- ▶ The core groupoid of $\Delta_{\perp}(C)$ is a discrete category with one object. Clearly $R\Delta_{\perp}(C)$ is not Morita equivalent to $R1$.
- ▶ Let $\Delta'_{\perp}(C)$ denote the five-element semigroup obtained by omitting ι .
- ▶ **NB** $R\Delta'_{\perp}(C)$ (with multiplication coming from the semigroup) is actually an R -algebra with identity element $\ell_C = \sigma_0 + \sigma_1 - v$.

A baby example with details, page 3

- Now in $R\Delta_{\perp}(C)$ we have a complete list of orthogonal idempotents:

$$\begin{aligned} e_0 &= \sigma_0\sigma_1, \quad e_1 = (1 - \sigma_0)\sigma_1, \\ e_2 &= (1 - \sigma_1)\sigma_0, \quad e_3 = (1 - \sigma_1)(1 - \sigma_0), \end{aligned}$$

so that, in the “Cauchy completion” $\mathcal{Q}R\Delta_{\perp}(C)$ (freely split idempotents and freely adjoin direct sums), we have

$$C \cong E_0 \oplus E_1 \oplus E_2 \oplus E_3$$

where E_i is the object obtained in splitting the idempotent e_i .

- If we put $\gamma = \sigma_1\sigma_0 - \sigma_0\sigma_1$ and $\delta = \tau - \sigma_1$, using the relations in the presentation, we obtain the equations

$$e_1\gamma = \gamma = \gamma e_2, \quad \delta e_1 = \delta = e_2\delta, \quad \gamma\delta = e_1, \quad \delta\gamma = e_2$$

so $\gamma \vdash: E_1 \cong E_2$. It follows that we have an isomorphism of the form

$$C \cong D \oplus E \oplus E \oplus F.$$

A baby example with details, page 4

- ▶ Transporting the endomorphisms σ_0, σ_1, τ on C across this isomorphism yields the matrices

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- ▶ Any endomorphism f of $D \oplus E \oplus E \oplus F$ which commutes with these three matrices must be of the form $f = f_0 \oplus f_1 \oplus f_1 \oplus f_2$.
- ▶ It follows that $R\Delta_{\perp}(C)$ is $\mathcal{V}$ -Morita equivalent to $R3$ where 3 is the discrete category with three objects. In other words, we have an equivalence of R -linear categories

$$[\Delta_{\perp}(C), \mathcal{V}] \cong \mathcal{V} \times \mathcal{V} \times \mathcal{V}.$$

Down to business: Factorization systems

1. Let $\mathcal{F}$ be a category with a factorization system $(\mathcal{E}, \mathcal{M})$. That is, $\mathcal{E}$ and $\mathcal{M}$ are sets of morphisms of $\mathcal{F}$ satisfying
 - FS0. for w invertible, $mw \in \mathcal{M}$ if $m \in \mathcal{M}$, while $we \in \mathcal{E}$ if $e \in \mathcal{E}$;
 - FS1. if $mh = ke$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$ then there exists a unique ℓ with $\ell e = h$ and $m\ell = k$;
 - FS2. every morphism f factors as $f = me$ for some $m \in \mathcal{M}$ and $e \in \mathcal{E}$.
2. It follows that $\mathcal{E} \cap \mathcal{M}$ is the set of invertible morphisms and that $\mathcal{E}$ and $\mathcal{M}$ are both closed under composition in $\mathcal{F}$.
3. We identify $\mathcal{E}$, $\mathcal{M}$ and $\mathcal{G} := \mathcal{E} \cap \mathcal{M}$ with the subcategories of $\mathcal{F}$ containing all objects.
4. Write $\mathcal{M}'$, $\mathcal{E}'$, $\mathcal{G}'$ for the sets of morphisms of $\mathcal{F}$ **not** in $\mathcal{M}$, $\mathcal{E}$, $\mathcal{G}$, respectively.

Proper factorization systems

The factorization system is called *proper* when

FSP. every member of $\mathcal{E}$ is an epimorphism and every member of $\mathcal{M}$ is a monomorphism.

If $(\mathcal{E}, \mathcal{M})$ is a proper factorization system on $\mathcal{F}$ then $(\mathcal{M}, \mathcal{E})$ is a proper factorization system on $\mathcal{F}^{\text{op}}$.

Proposition

In a proper factorization system, if $hf \in \mathcal{M}$ then $f \in \mathcal{M}$.

Dually, if $fk \in \mathcal{E}$ then $f \in \mathcal{E}$.

The bicategory $\mathfrak{R} := \mathcal{V}\text{-Mod}$

Objects of $\mathfrak{R}$ are $\mathcal{V}$ -categories (= R -linear categories or “ R -algebras with several objects”).

Hom categories are the $\mathcal{V}$ -functor $\mathcal{V}$ -categories

$$\mathfrak{R}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}] ;$$

objects of these homs are called *modules from $\mathcal{A}$ to $\mathcal{B}$* .

These homcategories are all abelian R -linear categories so, within them, **exact sequences have meaning**.

Module composition

$$\mathfrak{R}(\mathcal{B}, \mathcal{C}) \otimes \mathfrak{R}(\mathcal{A}, \mathcal{B}) \xrightarrow{\circ} \mathfrak{R}(\mathcal{A}, \mathcal{C})$$

is defined by “ tensoring over $\mathcal{B}$ ”:

$$(K \circ H)(C, A) = \int^B H(B, A) \otimes K(C, B) .$$

More on $\mathfrak{R}$

- ▶ For $\mathcal{V}$ -functors $\mathcal{A} \xrightarrow{F} \mathcal{C} \xleftarrow{G} \mathcal{B}$, we have the module $\mathcal{C}(G, F) : \mathcal{A} \rightarrow \mathcal{B}$ with components $\mathcal{C}(G, F)(B, A) = \mathcal{C}(GB, FA)$.
- ▶ In particular, for each $\mathcal{V}$ -functor $F : \mathcal{A} \rightarrow \mathcal{B}$ we have the module $F_* = \mathcal{B}(1_{\mathcal{B}}, F) : \mathcal{A} \rightarrow \mathcal{B}$.
- ▶ Also, the module $F^* = \mathcal{B}(F, 1_{\mathcal{B}}) : \mathcal{B} \rightarrow \mathcal{A}$ provides a right adjoint $F_* \dashv F^*$ for F_* in $\mathfrak{R}$.
- ▶ If $\mathcal{B}$ is Cauchy complete then every module from $\mathcal{A}$ to $\mathcal{B}$ with a right adjoint in $\mathfrak{R}$ is isomorphic to F_* for some $\mathcal{V}$ -functor $F : \mathcal{A} \rightarrow \mathcal{B}$.

Lemma on retracts of adjunctions

The lemma will be applied in $\mathfrak{R}$. It says that, if we have an idempotent 2-morphism on a morphism f with an adjoint u , then the splitting of that idempotent on f is a morphism with an adjoint obtained by splitting the mate idempotent on u . The form of the counit and unit is important. (A string proof is quite appealing.)

Lemma

In any bicategory $\mathfrak{M}$, suppose $f \dashv u : A \rightarrow X$ is an adjunction with counit $\varepsilon : fu \Rightarrow 1_A$ and unit $\eta : 1_X \Rightarrow uf$. Suppose $\omega : f \Rightarrow f$ is an idempotent 2-morphism on f with splitting provided by $\sigma : f \Rightarrow g$ and $\rho : g \Rightarrow f$: that is, $\rho\sigma = \omega$ and $\sigma\rho = 1_g$. The mate $\tilde{\omega} : u \Rightarrow u$ of $\omega : g \Rightarrow g$ under $f \dashv u$ is an idempotent 2-morphism on g satisfying $\varepsilon(\omega u) = \varepsilon(f\tilde{\omega})$. Then, any splitting of $\tilde{\omega}$ delivers a right adjoint v for g . Explicitly, if $\tilde{\rho}\tilde{\sigma} = \tilde{\omega}$ and $\tilde{\sigma}\tilde{\rho} = 1_v$ then $\tilde{\varepsilon} = \varepsilon(\rho\tilde{\rho})$ and $\tilde{\eta} = (\tilde{\sigma}\sigma)\eta$ provide a counit and unit for $g \dashv v$.

Modules as kernels of functorial transformations

- ▶ Every module $H : \mathcal{A} \rightarrow \mathcal{B}$ is the “kernel” of a functor $\hat{H} : [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$ defined by composition with H in $\mathfrak{K}$ thus:
 $\hat{H}(T) = (\mathcal{A} \xrightarrow{H} \mathcal{B} \xrightarrow{T} R\mathbf{1})$. That is,

$$\hat{H}(T)A = \int^B H(B, A) \otimes TB . \quad (1)$$

- ▶ We obtain a pseudofunctor $\widehat{(-)} : \mathfrak{K}^{\text{op}} \rightarrow \mathcal{V}\text{-Cat}$.
- ▶ Moreover, $\hat{H}$ has a right adjoint $\tilde{H} : [\mathcal{A}, \mathcal{V}] \rightarrow [\mathcal{B}, \mathcal{V}]$ given by right extension along H in $\mathfrak{K}$. Explicitly,

$$\tilde{H}(F)B = \int_A [H(B, A), FA] = [\mathcal{A}, \mathcal{V}](H(B, -), F) . \quad (2)$$

The inclusion $j : R\mathcal{G} \hookrightarrow R\mathcal{F}$ gives the right adjoint $\mathcal{V}$ -module $j^* : R\mathcal{F} \rightarrow R\mathcal{G}$ where $j^*(A, B) = R\mathcal{F}(jA, B)$.

Lemma

For $g \in \mathcal{G}(C, A)$ and $h \in \mathcal{F}(B, D)$, there is a commutative square

$$\begin{array}{ccc}
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B) \\
 \downarrow & & \downarrow R\mathcal{F}(g, h) \\
 R\mathcal{M}'(C, D) & \xrightarrow{\subseteq} & R\mathcal{F}(C, D)
 \end{array} \tag{3}$$

so that $M'(A, B) = R\mathcal{M}'(A, B)$ defines a submodule M' of j^* . There is a short exact sequence

$$0 \longrightarrow M' \xrightarrow{\subseteq} j^* \xrightarrow{q} M \longrightarrow 0 \tag{4}$$

in $\mathfrak{R}(R\mathcal{F}, R\mathcal{G})$ where $M(A, B) = R\mathcal{M}(jA, B)$.

Dually, there is a short exact sequence

$$0 \longrightarrow E' \xrightarrow{\subseteq} j_* \xrightarrow{\tilde{q}} E \longrightarrow 0 \quad (5)$$

in $\mathfrak{K}(R\mathcal{G}, R\mathcal{F})$ where $E(B, A) = R\mathcal{C}(B, jA)$.

So now we have modules in both directions

$$M : R\mathcal{F} \rightarrow R\mathcal{G} \quad \text{and} \quad E : R\mathcal{G} \rightarrow R\mathcal{F} .$$

Do they give an equivalence in $\mathfrak{K}$?

Not in general, however:

$$M \circ E \xrightarrow{\cong} 1_{R\mathcal{G}}$$

Lemma

The family of R -module morphisms

$$\phi_B : E(B, D) \otimes M(C, B) \rightarrow R\mathcal{G}(C, D) ,$$

defined by, for $e \in \mathcal{E}(B, D)$ and $m \in \mathcal{M}(C, B)$,

$$\phi_B(e \otimes m) = \begin{cases} em & \text{for } em \in \mathcal{G} \\ 0 & \text{otherwise} \end{cases}, \text{ is dinatural in } B \in \mathcal{F}, \text{ natural in}$$

$C, D \in \mathcal{G}$, and induces an isomorphism

$$\bar{\phi} : M \circ E \xrightarrow{\cong} 1_{R\mathcal{G}}$$

in $\mathfrak{K}$ which gives an isomorphism $\hat{E} \circ \hat{M} \cong 1_{[\mathcal{G}, \mathcal{V}]}$. The mate of this isomorphism, under the adjunction $\hat{E} \dashv \tilde{E}$, yields a natural transformation

$$\Theta : \hat{M} \rightarrow \tilde{E} .$$

There are alternative formulas for $\hat{M}$ and $\tilde{E}$ which can be useful. For $A \in \mathcal{F}$, let $\text{Sub}A$ and $\text{Quo}A$ denote representative sets of the isomorphism classes of $\mathcal{M}/A$ and $A/\mathcal{C}$, respectively. Then

$$\hat{M}(T)A \cong \sum_{(U \xrightarrow{m} A) \in \text{Sub}A} TU \quad \text{and} \quad \tilde{E}(T)A \cong \prod_{(A \xrightarrow{e} W) \in \text{Quo}A} TW. \quad (6)$$

In cases where $\text{Sub}A$ and $\text{Quo}A$ are finite, both the sum and product in these alternative formulas are direct sums and the components of $\Theta : \hat{M} \rightarrow \tilde{E}$ transport across the isomorphisms to matrices whose non-zero entries have the form $T(em) : TU \rightarrow TW$ for $em \in \mathcal{G}$. Suppose that elements of each $\text{Sub}A$ can be listed $m_0, \dots, m_s$ and those of each $\text{Quo}A$ can be listed $e_0, \dots, e_s$, in such a way that $e_i m_j \in \mathcal{G}$ implies $1 \leq i \leq j \leq s$ and that $e_i m_i = 1$ for $0 \leq i \leq s$. Then these matrices are square and triangular with identities down the main diagonal, and so (because we have additive inverses in the hom R -modules) are invertible.

Example

Let $\mathcal{A}$ be any category in which every morphism is a monomorphism, pullbacks exist, and each slice category $\mathcal{A}/A$ has finitely many isomorphism classes. Take $\mathcal{F} = \mathcal{A}\sharp$ to be the category of spans in $\mathcal{A}$; that is, the objects are those of $\mathcal{A}$ and the morphisms $f : A \rightarrow B$ are isomorphism classes $[f_1, U, f_2]$ of spans $A \xleftarrow{f_1} U \xrightarrow{f_2} B$ in $\mathcal{A}$, where composition uses pullback. Take $\mathcal{M}$ to consist of those spans f with f_1 invertible and $\mathcal{E}$ to consist of those spans f with f_2 invertible. The core groupoid $\mathcal{G}$ of $\mathcal{F}$ is isomorphic to the core groupoid of $\mathcal{A}$. Here $\hat{M}$ is an equivalence; the Lack-St (2015) proof uses a comonadicity argument. An example of this is $\mathcal{A} = \mathbf{FI}$, the category of finite sets and injective functions; so $\mathcal{F} = \mathbf{FI}\sharp$. Then $\mathcal{G} = \mathfrak{S}$, the groupoid of finite sets and bijections, called *the symmetric groupoid*: so $[\mathcal{G}, \mathcal{V}]$ is the category of R -linear Joyal species; it is equivalent to the product over $n \geq 0$ of the categories of R -linear representations of the groups $\mathfrak{S}_n$. In this case, the deduced equivalence $[\mathbf{FI}\sharp, \mathcal{V}] \simeq [\mathfrak{S}, \mathcal{V}]$ is in Teimuraz Pirashvili (2000).

Plan to apply lemma on retracts of adjunctions

We continue with a proper factorization system $(\mathcal{E}, \mathcal{M})$ on a category $\mathcal{F}$. The goal is to find other conditions under which $M : R\mathcal{F} \rightarrow R\mathcal{G}$ in $\mathfrak{A}$ is an equivalence. If this is to be the case, M will need to have a left adjoint so we would like to apply our Lemma to deduce the adjoint from the adjunction $j_* \dashv j^*$. This means we would like M to be a retract of j^* . A natural splitting of the short exact sequence (4) would suffice. A splitting $\rho : M \rightarrow j^*$ of the epimorphism $q : j^* \rightarrow M$ has components

$$\rho_{A,B} : M(A, B) \rightarrow R\mathcal{F}(A, B)$$

which are natural in $A \in \mathcal{G}$ and $B \in \mathcal{F}$. In particular they are natural in $B \in R\mathcal{M}$ and so, by Yoneda, are determined by the value, say $\rho_A \in R\mathcal{F}(A, A)$, at $1_A \in R\mathcal{M}(A, A)$. Indeed $\rho_{A,B}(m) = m\rho_A$. This motivates the following assumption.

Idempotent Axiom. For each object $A \in \mathcal{F}$, there is a morphism $p_A : A \rightarrow A$ in $R\mathcal{F}$ such that

- p0. $p_A p_A = p_A$;
- p1. if $f \in \mathcal{F}(A, B)$ and $f \in \mathcal{M}'$ then $f p_A = 0$;
- p2. if $f \in \mathcal{F}(A, B)$ and $f \in \mathcal{E}'$ then $p_B f = 0$;
- p3. if $g \in \mathcal{G}(A, B)$ then $g p_A = p_B g$;
- p4. $p'_A = 1_A - p_A \in R(\mathcal{E}'(A, A) \cap \mathcal{M}'(A, A))$.

Example

Every groupoid $\mathcal{G}$ provides a trivial example with $\mathcal{F} = \mathcal{E} = \mathcal{M} = \mathcal{G}$ and $p_A = 1_A$. In the general case, we can think of $p'_A = 1_A - p_A$ as an obstruction to $\mathcal{F}$'s being a groupoid although we will see below the sense in which $\mathcal{F}$ is not too far from $\mathcal{G}$ at the R -linear level.

The retraction σ

Using p1, we have the identity linear function as the left side of a commutative square (7). Since the factorization system is proper, if $u \in \mathcal{M}'(A, A)$ and $f \in \mathcal{F}(A, B)$ then $fu \in \mathcal{M}'$; so, by p4, $fp'_A \in R\mathcal{M}'(A, B)$ for all $f \in \mathcal{F}(A, B)$. This gives a splitting $\sigma_{A,B}$ of the idempotent $R\mathcal{F}(p'_A, 1_B)$.

$$\begin{array}{ccc}
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B) \\
 \downarrow 1 & \swarrow \sigma_{A,B} & \downarrow R\mathcal{F}(p'_A, 1_B) \\
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B)
 \end{array} \tag{7}$$

Using p3, we see that the linear functions $R\mathcal{F}(p'_A, 1_B)$ are the components of an idempotent module morphism $\pi' : j^* \rightarrow j^*$ which is split by the module morphism $\sigma : j^* \rightarrow M'$ and the inclusion $M' \hookrightarrow j^*$. It follows that the short exact sequence for M splits so that, if we put $\pi = 1 - \pi' = R\mathcal{F}(p_A, 1_B)$, we have:

Lemma

The idempotent module morphism $\pi = R\mathcal{F}(p_A, 1_B) : j^* \rightarrow j^*$ splits as

$$\begin{array}{ccc}
 j^* & \xrightarrow{\pi} & j^* \\
 \searrow q & & \nearrow \rho \\
 & M & \\
 & \xrightarrow{1_M} & M \\
 & \nearrow q & \searrow q
 \end{array}
 \quad (8)$$

where, for $m \in \mathcal{M}(A, B)$, $\rho_{A,B}(m) = mp_A$ and for $g \in \mathcal{G}(A', A)$ and $f \in \mathcal{F}(B, B')$,

$$M(g, f)m = fmgp_{A'}.$$

Dually, we also have:

Lemma

The idempotent module morphism $\varpi = R\mathcal{F}(1_A, p_B) : j_* \rightarrow j_*$ splits as

$$\begin{array}{ccccc}
 j_* & \xrightarrow{\varpi} & j_* & & \\
 & \searrow \tilde{q} & \nearrow \tilde{\rho} & \searrow \tilde{q} & \\
 & & E & \xrightarrow{1_E} & E
 \end{array} \tag{9}$$

where $E(A, B) = R\mathcal{E}(A, B)$, $\tilde{\rho}_{A,B}(e) = p_B e$, and

$$E(f, g)e = p_{B'} g e f ,$$

for $e \in \mathcal{E}(A, B)$, $f \in \mathcal{F}(A', A)$, $g \in \mathcal{E}((B, B'))$.

Mates splitting

Lemma

The idempotent module morphisms $\pi : j^ \rightarrow j^*$ and $\varpi : j_* \rightarrow j_*$ are mates under the adjunction $j_* \dashv j^*$ in $\mathfrak{K}$. By the retract-of-adjunction Lemma, $E \dashv M$ in $\mathfrak{K}$.*

Since $\widehat{(-)} : \mathfrak{K}^{\text{op}} \rightarrow \mathcal{V}\text{-Cat}$ and $\widetilde{(-)} : \mathfrak{K}^{\text{co}} \rightarrow \mathcal{V}\text{-Cat}$ are pseudofunctors:

Corollary

The functor $\widehat{M} : [\mathcal{G}, \mathcal{V}] \rightarrow [\mathcal{F}, \mathcal{V}]$ has right adjoint $\widehat{E}$. The functor $\widetilde{M} : [\mathcal{F}, \mathcal{V}] \rightarrow [\mathcal{G}, \mathcal{V}]$ has right adjoint $\widetilde{E}$. So $\widehat{E} \cong \widetilde{M}$.

Lemma

The unit $1_{R\mathcal{G}} \Rightarrow M \circ E$ of the adjunction $E \dashv M$ in $\mathfrak{X}$ is invertible. So the unit $1_{[\mathcal{G}, \mathcal{V}]} \Rightarrow \hat{E} \circ \hat{M}$ of $\hat{M} \dashv \hat{E}$ is invertible and $\hat{M}$ is fully faithful.

Proof.

Note that $(j^* \circ j_*)(C, D) \cong R\mathcal{F}(jC, jD)$ and, from the adjunction retract lemma, the unit of the adjunction $E \dashv M$ has component at (C, D) equal to the composite

$$\begin{aligned} R\mathcal{G}(C, D) &\xrightarrow{j} R\mathcal{F}(jC, jD) \cong \int^{B \in \mathcal{F}} R\mathcal{F}(B, jD) \otimes R\mathcal{F}(jC, B) \\ &\xrightarrow{\int^{B \in \mathcal{F}} q \otimes \tilde{q}} (M \circ E)(C, D) \end{aligned}$$

which is inverse to the component of our earlier isomorphism $\bar{\phi}$. □

The hardest bit

Lemma

If each object A of $\mathcal{F}$ has only finitely many $\mathcal{M}$ -subobjects (that is, each slice category $\mathcal{M}/A$ has only finitely many isomorphism classes) then the counit of the adjunction $E \dashv M$ in $\mathfrak{R}$ is a split epimorphism.

Proof.

The counit, which is natural, is the composite

$$\begin{aligned} (E \circ M)(A, B) &= \int^{C \in \mathcal{G}} R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C) \\ &\rightarrow (j_* \circ j^*)(A, B) \rightarrow R\mathcal{F}(A, B) \end{aligned}$$

which takes the equivalence class of $m \otimes e \in R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C)$ to $mp_C e \in R\mathcal{F}(A, B)$. We claim this family has a natural right inverse.

To be continued.



Proof continued.

In particular, for $A = B$, we must see that the identity 1_A of A is in the image. Take a finite family $(C_i \xrightarrow{m_i} A)_{i=0}^k$ of morphisms in $\mathcal{M}$ representing all isomorphism classes in the ordered set $\mathcal{M}/A$ and with the property that $m_i = m_j n$ for some $C_i \xrightarrow{n} C_j$ implies $i \leq j$; we can suppose $C_k = A$ and $m_k = 1_A$. Let

$$\phi_j : \bigoplus_{i \leq j} R\mathcal{M}(C_i, A) \otimes R\mathcal{E}(A, C_i) \longrightarrow R\mathcal{F}(A, A)$$

denote the function defined by $\phi_j(m \otimes e) = mp_{C_i}e$ for $m \in R\mathcal{M}(C_i, A)$ and $e \in R\mathcal{E}(A, C_i)$. So the image of ϕ_i is contained in the image of ϕ_j for $i \leq j$. By induction on j one can show that every $f \in \mathcal{F}(A, A)$ is in the image of the function ϕ_k .

Hence, $1_A \in \mathcal{F}(A, A)$ is in the image of the A, A component of the counit.

To be continued.



Proof continued further.

Suppose $t \in \int^{C \in \mathcal{G}} R\mathcal{M}(C, A) \otimes R\mathcal{E}(A, C)$ maps to $1_A \in \mathcal{F}(A, A)$. By the Yoneda Lemma, there exists a unique family of morphisms

$$R\mathcal{F}(A, B) \longrightarrow \int^{C \in \mathcal{G}} R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C)$$

which is natural in $B \in \mathcal{F}$ taking $1_A \in \mathcal{F}(A, A)$ to t . Again by Yoneda, this gives right inverses to the components of the counit. Dually, there is such a family natural in B . By Yoneda yet again, these families agree since they have the same value at the identity. So the counit of $E \dashv M$ has a natural right inverse.

That's really the end of proof!



Main Theorem

Theorem

If each object A of $\mathcal{F}$ has only finitely many $\mathcal{M}$ -subobjects then the adjunction $E \dashv M$ is an equivalence $R\mathcal{F} \simeq R\mathcal{G}$ in $\mathfrak{R}$.

Proof.

We now know that the unit $1_{R\mathcal{G}} \Rightarrow \tilde{M} \circ \hat{M}$ is invertible. So $\hat{M} : [R\mathcal{G}, \mathcal{V}] \rightarrow [R\mathcal{F}, \mathcal{V}]$ is fully faithful and so is its composite $M' : R\mathcal{G}^{\text{op}} \rightarrow [R\mathcal{F}, \mathcal{V}]$, $C \mapsto M(C, -)$, with the Yoneda embedding. Moreover, one of our Lemmas implies M' is strongly generating. By Theorem 2 of Day-St (1986, about strong generators being dense), M' is also dense. So the counit of $E \dashv M$ is invertible. □

In the terminology of ring theory, this Theorem implies that $R\mathcal{F}$ and $R\mathcal{G}$ are Morita equivalent several-object R -algebras. In the terminology of enriched category theory, it implies that $R\mathcal{F}$ and $R\mathcal{G}$ are Cauchy equivalent $\mathcal{V}$ -categories.

Consequences

Corollary

*The functor $\widehat{M} : [\mathcal{G}, \mathcal{V}] \rightarrow [\mathcal{F}, \mathcal{V}]$ is an equivalence with pseudo-inverse $\widehat{E}$.
The functor $\widetilde{M} : [\mathcal{F}, \mathcal{V}] \rightarrow [\mathcal{G}, \mathcal{V}]$ is an equivalence with pseudo-inverse $\widetilde{E}$.*

Corollary

The equivalence $\widetilde{M}$ is a retract of the restriction functor

$$[j, 1] : [R\mathcal{F}, \mathcal{V}] \rightarrow [R\mathcal{G}, \mathcal{V}] .$$

Corollary

For any R -linear category $\mathcal{X}$ in which idempotents split and finite direct sums exist, there is an equivalence

$$[\mathcal{F}, \mathcal{X}] \simeq [\mathcal{G}, \mathcal{X}] .$$

Stiffness

Definition

In any category $\mathcal{F}$, a morphism $f : A \rightarrow B$ is called *stiff* when the only endomorphisms of A through which f factors are automorphisms. In other words, $f = (A \xrightarrow{u} A \xrightarrow{v} B)$ implies u invertible. A morphism is *costiff* when it is stiff in the opposite category. A category is called *stiff* when the costiff and stiff morphisms are the $\mathcal{E}$ and $\mathcal{M}$ of a proper factorization system.

The category of finite sets and the category of finite-dimensional vector spaces over a field are stiff categories: costiff = surjective, stiff = injective. Any stiff endomorphism is an automorphism so a consequence of stiffness is the **pigeon-hole principle**:

Proposition (PhP)

For all objects A in a stiff category $\mathcal{F}$, the inclusions $\mathcal{M}(A, A) \subseteq \mathcal{G}(A, A)$ and $\mathcal{E}(A, A) \subseteq \mathcal{G}(A, A)$ are equalities. Equivalently, if $\mathcal{G}(A, B) \neq \emptyset$ then $\mathcal{G}(A, B) = \mathcal{E}(A, B) = \mathcal{M}(A, B)$.

Some work of Laci Kovács

Recall that the set $\mathcal{G}'$ of morphisms is the complement of $\mathcal{G}$ in $\mathcal{F}$. For each $C \in \mathcal{F}$, **PhP** implies that $R\mathcal{G}'(C, C)$ is a two-sided ideal in $R\mathcal{F}(C, C)$. In particular $R\mathcal{G}'(C, C)$ is an R -algebra, possibly without an identity element.

In the case where $\mathcal{F} = \text{vect}_{\mathbb{F}}$ is the category of finite vector spaces over a finite field $\mathbb{F}$ and the factorization is surjective-injective $\mathbb{F}$ -linear functions, Laci Kovács (1992) produced an identity element making $R\mathcal{G}'(C, C)$ a unital algebra **provided the characteristic of $\mathbb{F}$ is invertible in R** ; also see Subsection 3.1 of Kuhn's paper for helpful material on this.

Therefore we make the following definition in the general situation.

Definition

A *Kovács idempotent* is an identity element ℓ_C making $R\mathcal{G}'(C, C)$ an R -algebra under composition.

From Kovács to the Idempotent Axiom

Lemma

Assume *PhP* and that each object C has a Kovács idempotent ℓ_C . Then ℓ_C is a central idempotent in the R -algebra $R\mathcal{F}(C, C)$ and the morphisms $\ell_C : C \rightarrow C$ are the components of a natural endomorphism of the inclusion functor of $j : R\mathcal{G} \rightarrow R\mathcal{F}$.

Proposition

Assume that $\mathcal{F}$ is stiff and that each object C has a Kovács idempotent ℓ_C . Then the idempotents $p_A = 1_A - \ell_A : A \rightarrow A$ satisfy the Idempotent Axiom.

Semisimplicity

It can happen that $R\mathcal{F}(C)$ is semisimple. This is true when $Y = \mathcal{F}(C)$ is any finite monoid of Lie type and R is a field of characteristic 0 [Okniński-Putcha (1991)]. Steinberg's 2016 book provides iff conditions on a finite monoid Y and field R in order for RY to be semisimple; in particular, the characteristic of R should not divide the order of the group of invertible elements in the monoid pYp for any idempotent $p \in Y$. Itamar Stein has proved all the endomorphism monoids $R\Delta_{\perp}(\mathbf{a})$ of $R\Delta_{\perp}$ are semisimple for R any field.

With semisimplicity of RY , the inclusion of every two-sided ideal J of RY into RY splits as a left module morphism and splits as a right module morphism. The value of the left module splitting at the identity of RY gives a right identity for J and the value of the right module splitting at the identity of RY gives a left identity for J . So J becomes an R -algebra with identity. This is a source of Kovács idempotents for categories $\mathcal{F}$ with endomorphism monoids $\mathcal{F}(C)$ finite and $R\mathcal{F}(C)$ semisimple.

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Thank you for your attention 😊

